Research Talk: Statistical Methodology for Real-World ML Systems Monitoring

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Research

It mainly focus on developing rigorous, computational efficient and generalizable Statistical Methodology to understand the change of data - specifically in the deployed Machine Learning(ML) systems. Main Scientific & Practical Questions to address:

- Given two batches of data $D_{exist} = (X_{exist}, Y_{exist})$ and $D_{new} = (X_{new}, Y_{new})$. How to know whether the difference is from concept drift ($P(Y|X)$) or covariate shift($P(X)$)?
- How to understand the change of data from the feature level - how to do trustworthy attribution?
- Given a deployed ML model in the production system, how to know when and how should we update it?
- After identifying the model performance drop, how to adapt them with high interpretability and low latency?

Overview of Methods

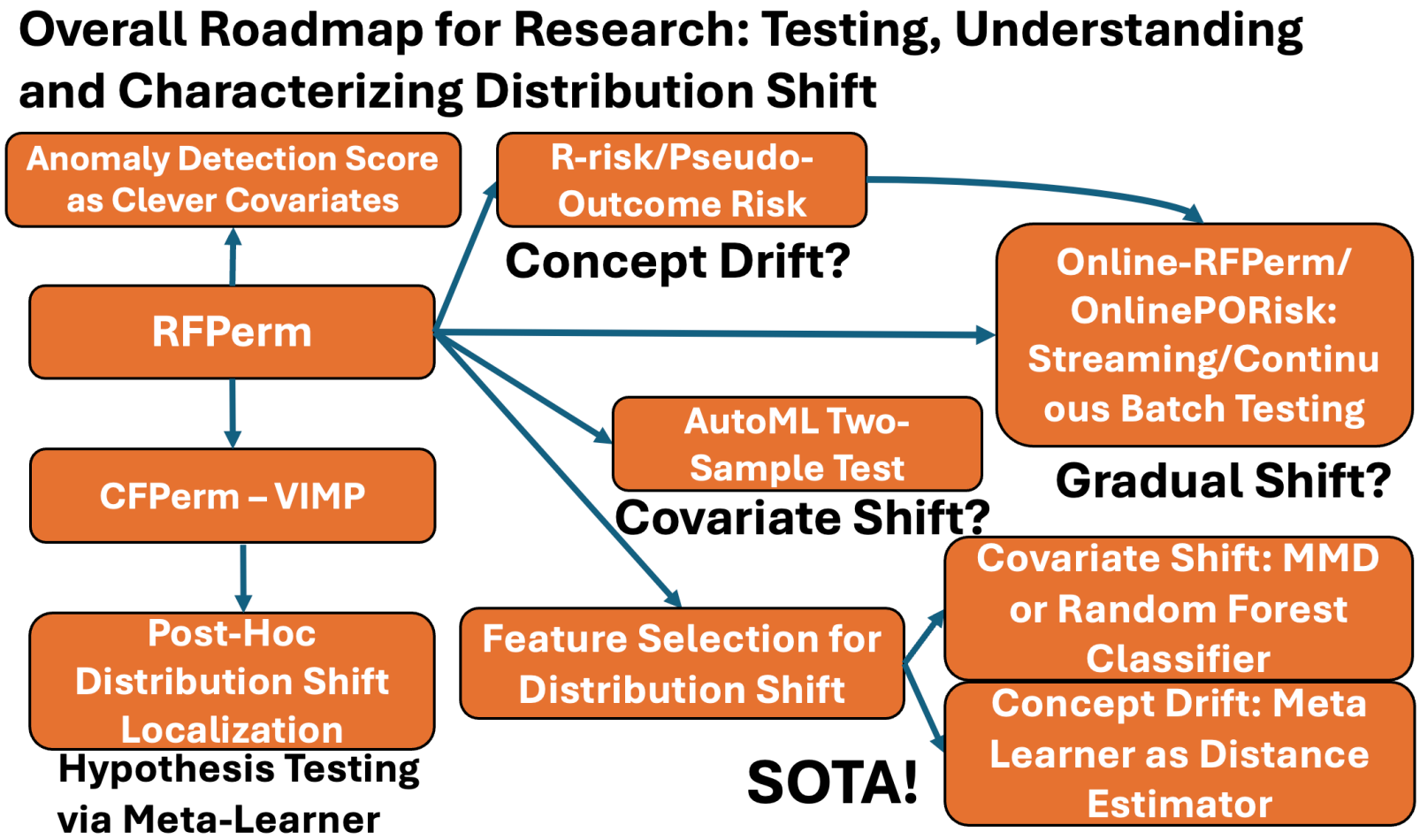


Figure: Overview of the Methodologies we have developed - understanding and adapting distribution shift

Motivation for the Methodology - Impossibility Issue

- Decomposing the observed performance degradation into separate contributions from model shift and data distribution shift is impossible without additional structural assumptions.
- Attributing the performance gap uniquely to either covariate shift or concept drift is impossible from the observed error alone, as the two effects are observationally equivalent under standard supervised learning setups.
- In the observed distribution shift, disentangle the shift(covariate shift or concept drift) into individual dimension is impossible.
- Thus developing hypothesis testing procedure that can detect both types of the shifts would be highly desirable for practical value - both robust and efficient.

Methodology 1 - Random Forest Permutation Test

Permutation test to compare validation error(OOB Error) and test error(Prediction Error) in Random Forest.

- Random Forest, well known for its robustness across a wide variety of different scenarios, would be the best choice to isolate the effect of OOD(out-of-distribution).
- It would be capable of characterizing both Concept Drift and Covariate Shift – Density Aware nature of Random Forest.
- Generalizable to arbitrary Machine Learning Models on arbitrary Modality of data - as long as the validation set is well-defined. The software is open-sourced in https://github.com/ruanhq-StatsML/RFPerm.

Methodology 2 - Meta-Learner based Hypothesis Testing

Leverage Meta-Learners to detect Distribution Shift - via Variable Importance. D_{exist} as Control Batch($T = 0$), D_{new} as Treatment Batch($T = 1$).

- Permute-then-refit the Causal Learner, looking at the variable importance(LOCO or permutation variable importance).
- The variable importance in this framework itself can be interpreted as the quantification of the contribution to certain types of distribution shift, potentially OOD variable importance.
- The meta-learner serves as a proxy as the distance estimator instead of causal estimator. The software is open-sourced in https://github.com/ruanhq-StatsML/Causal_Objective_Permutation_Test.

Meta-Learner Testing for Feature Selection and Distribution Shift Localization

The proposed meta-learner hypothesis testing framework (CFPerm) inherently serves a dual purpose:

1. Identifying informative features driving distributional changes.
2. Enabling subset localization of the sources of shift.

- Decision Rule: A feature is flagged as a significant driver if its original Variable Importance (VIMP) score is significantly larger than its permuted VIMP distribution under the null hypothesis.
- Subset Localization: The selected significant features not only reveal the primary drivers of the distribution shift, but the distinct subsets they induce also serve as highly efficient proxies for pinpointing where the shift is most pronounced.
  - Insight: This facilitates interpretable diagnostics by directly targeting high-drift regions in the feature space.

Methodology 3 - R-learner based Hypothesis Testing

Leverage Meta-Learners, R-learner to develop hypothesis testing procedure for isolating concept drift from covariate shift(Causal objetive function)

- The motivation of R-learner is similar to that in the Double/Debiased Machine Learning(DML) - Remove the effect of the covariates.
- Based on Projection Theory, $\tilde{Y} = Y - m(X) = (I - \Pi_X)(Y)$ and $\tilde{T} = T - e(X) = (I - \Pi_X)T$. The R-risk is invariant under Covariate Shift, isolating of the Concept Drift, serve as an objective function for isolating the concept drift.
- Can be generalizable to arbitrary Objective Functions in literatures of Causal Inference.

Methodology 3 - Feature Selection for Distribution Shift

In production, to conduct feature-level attribution:

- For Covariate Shift only, MMD+LOCO(leave-one-covariate-out) or train a random forest to distinguish two datasets is efficient.
- For Concept Drift, leveraging the Meta-Learner(DR-learner with Permutation VIMP) procedure for Feature Selection.
- Proactive and Actionable diagnostic tools for feature quality deterioration - feature selection in near real-time, enabling tractable root-cause analysis.

Methodology 4 - Product Consideration

- Under the KL-divergence, the distance between the joint distributions of existing and new data satisfies

$$\mathrm{D}(\mathrm{P}(\mathrm{X}_{\mathrm{exist}}, \mathrm{Y}_{\mathrm{exist}}), \mathrm{P}(\mathrm{X}_{\mathrm{new}}, \mathrm{Y}_{\mathrm{new}})) =$$

$\mathrm{D}(\mathrm{P}(\mathrm{Y}_{\mathrm{new}}|\mathrm{X}_{\mathrm{new}}), \mathrm{P}(\mathrm{Y}_{\mathrm{exist}}|\mathrm{X}_{\mathrm{exist}})) +$
$\mathrm{D}(\mathrm{P}(\mathrm{X}_{\mathrm{new}}), \mathrm{P}(\mathrm{X}_{\mathrm{exist}}))$.(1) However, this decomposition property holds exclusively for KL-divergence, which precludes its direct application to disentangle the overall data change into covariate shift and concept drift.

- In the event of a significant performance drop in the deployed model, adopting a post-hoc localization approach for the distribution shift is often more practical than attempting to explicitly decompose the shift into its constituent sources.

Methodology 4 - Online Extension of RFPerm

Suppose a deployed model $\hat{f}$ is launched in Production. In the streaming data setting, new observations $\cup_{i=1}^{n}\{X_i, Y_i\}$ coming continuously, we aim to give people an idea of when the gradual distribution shift become significant - trigger model update.

- A procedure based on empirical p-values is proposed to assess the extremity of the current predictive error relative to those in the deployed model.
- An online False Discovery Rate (FDR) control procedure is incorporated to effectively mitigate false alarms.
- Model Agnostic as long as a valid predictive model is well-defined and near-real-time inference is enabled.
- The software is available at https://github.com/ruanhq-StatsML/OnlineRFPerm.

Methodology 4: Production Value - adapt distribution shift

The score for the OnlineRFPerm procedure can serve as a clever covariate for the model update procedure which will help alleviating the drop of the MSE introduced by the distribution shift.

- The clever covariate can serve as an optional fallback procedure to enhance the next stage of the predictive modeling - under no distribution shift, this feature will not yield significance in predicting the responses.
- Additionally, examining the confidence intervals associated with its variable importance offers practical insights into how this feature behaves across continuously arriving batches of observations - via the OnlinePFI(online permutation feature importance) procedures.

Methodology 4: Product Value

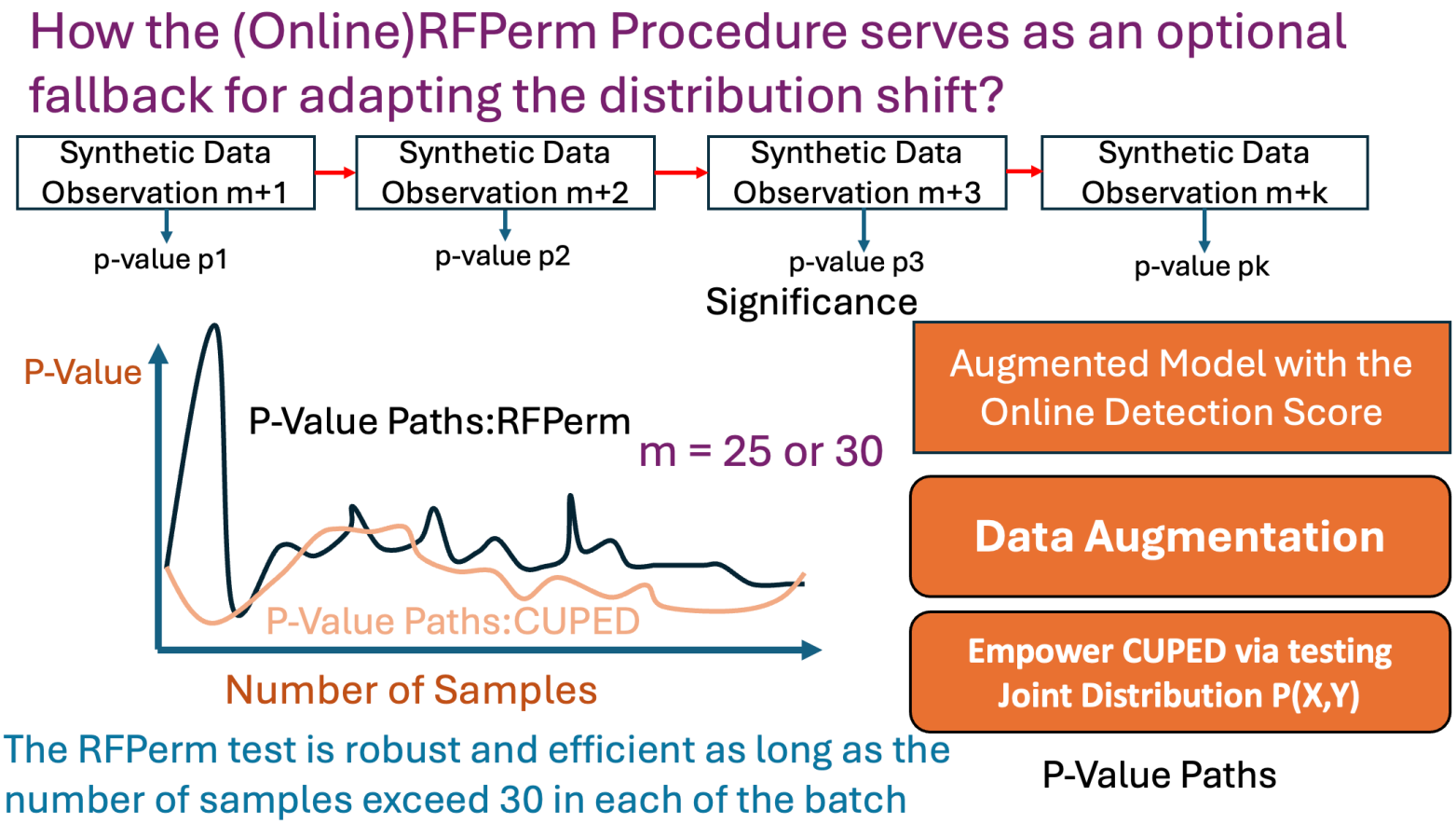


Figure: Overview of the Product Landing Value for this RFPerm procedure

Methodology 4: Illustration

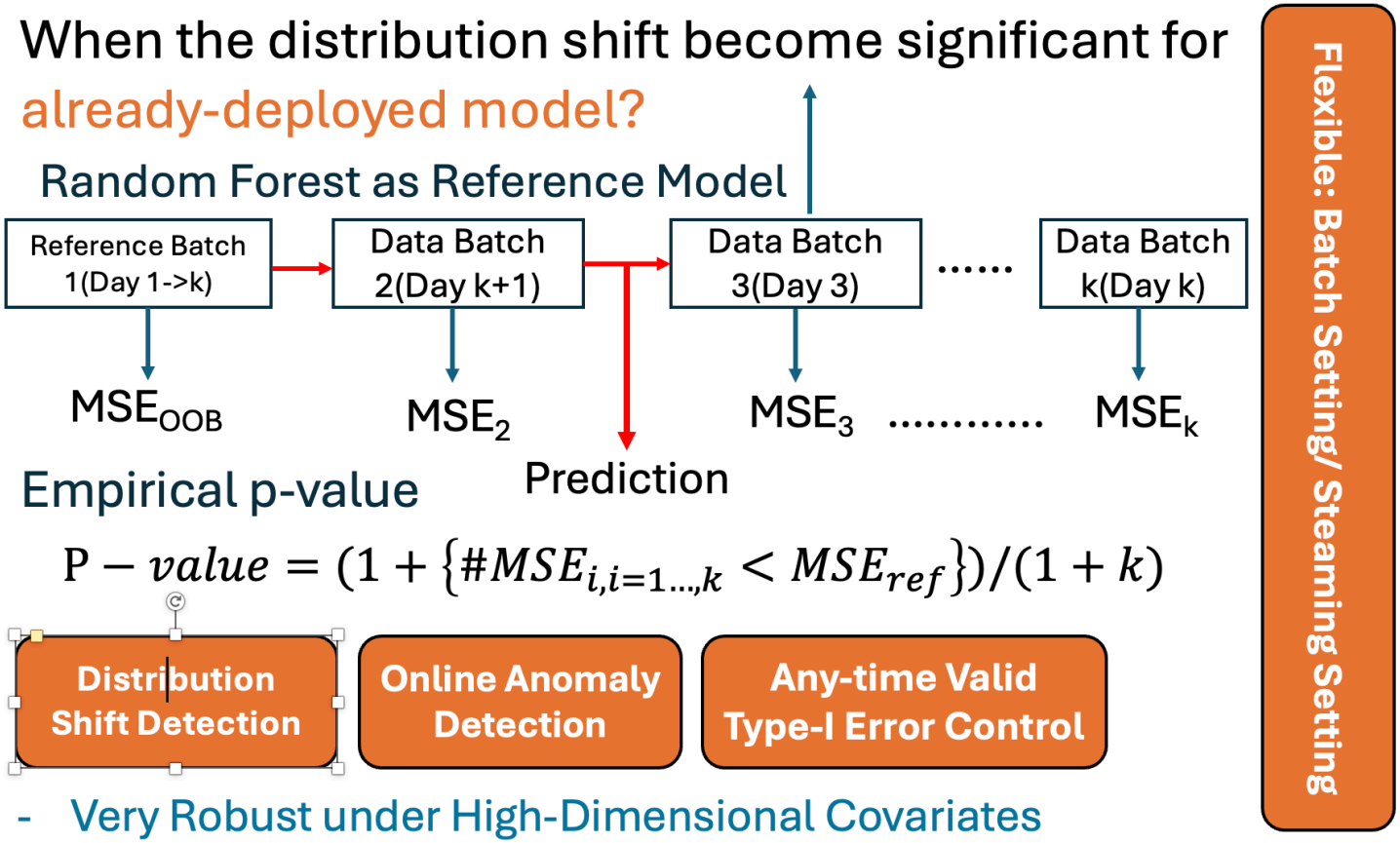
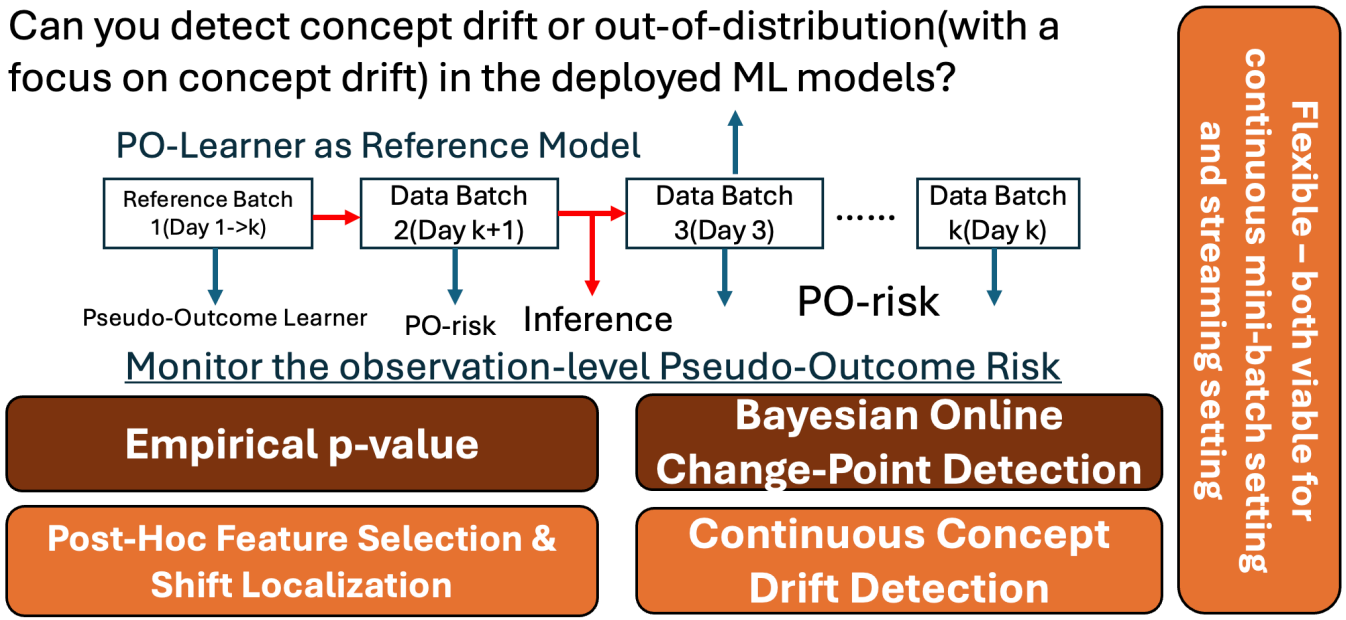


Figure: Streaming Testing for Gradual Distribution

Methodology 4: Illustration



- Not Sensitive to feature dimensions, PO-risk as Distance Estimator
- Streaming Concept Drift Detection via Monitoring Distance
- Flexibility for the component model.
- Is the model performance drop comes from the concept drift or covariate shift?

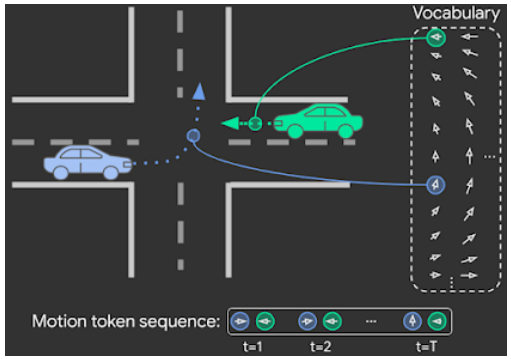
It can be extended to detect the concept drift in the continuous time concept drift detection - via pseudo-outcome learner as distance-estimator - quantify the distribution shift continuously.

Formal Extension: Monitoring the PO-risk for OOD

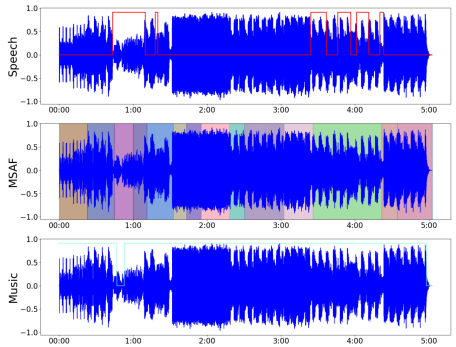
- Leverage pseudo-outcome (PO) risk for continuous, streaming monitoring of the degree of distribution shift—enabling real-time OOD tracking.
- Introduce the first-ever methodology to quantify uncertainty in OOD estimation through online bootstrap confidence intervals for PO-risk.
- Provide a readily deployable product dashboard for OOD monitoring with formal uncertainty quantification.

Methodology 4: Flexibility and Modality Agnostic

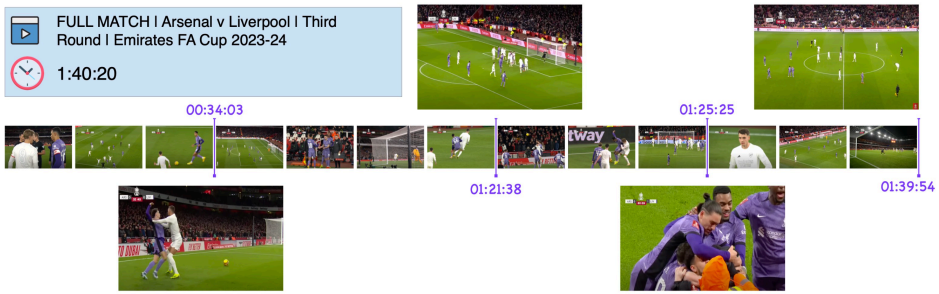
The methodology is model-agnostic, with demonstrated effectiveness across diverse data modalities—making it broadly applicable to any ML system without architectural modification.



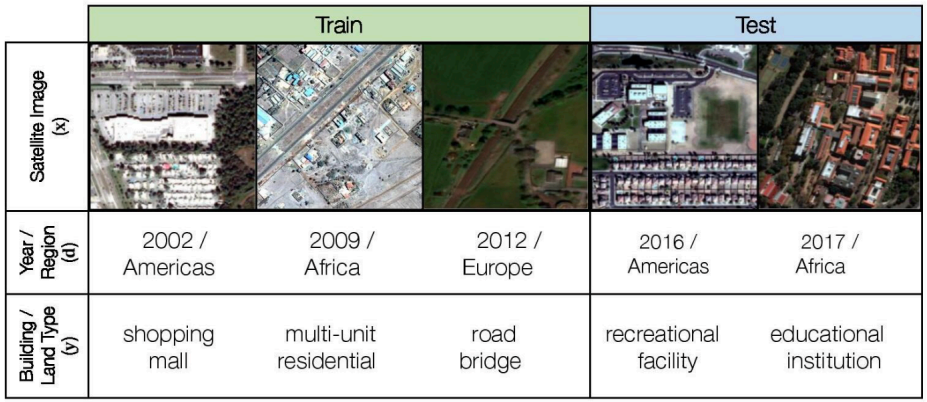
(a) Motion Planning



(b) Audio



(c) Video Streaming



(d) Functional Map (Image)

Feature Selection for Distribution Shift

- Given two batches of data, how should one know how the shift can be explained by the individual features? It can be formulated as a Feature Selection problem for the Distribution Shift(both covariate shift and concept drift).
- It's a problem that every data scientist will encounter with real-world scenarios, tractable practical values as well as the promising extensions build upon the current Statistical & Machine Learning Methodologies.
- This procedure can be embedded in the deployed ML systems - along with the existing practices including KL-divergence & PSI.
- The post-hoc localization procedures are also available to quantify the subset-level distance(MMD, PO-risk).

Feature Selection for Covariate Shift

We want to understand the difference of $P(X_{exist})$ and $P(X_{new})$ - which subset of features are contributing significantly to the covariate shift?

- MMD-LOCO: Maximum Mean Discrepancy(MMD) equipped with the leave-one-covariate procedure - calculate the difference as the ranking mechanism for the feature selection.
- RF-Domain Classifier: A binary classifier is trained on distinguishing the two batches of datasets X_{exist} and X_{new}, borrowing the ideas from the C2ST(Classifier Two-Sample Testing Procedure) and regard the variable importance ranking for the feature selection criterion.
- The random forest based procedure is highly competitive in comparison to the existing SOTA - FSL-Net.

Feature Selection for Concept Drift

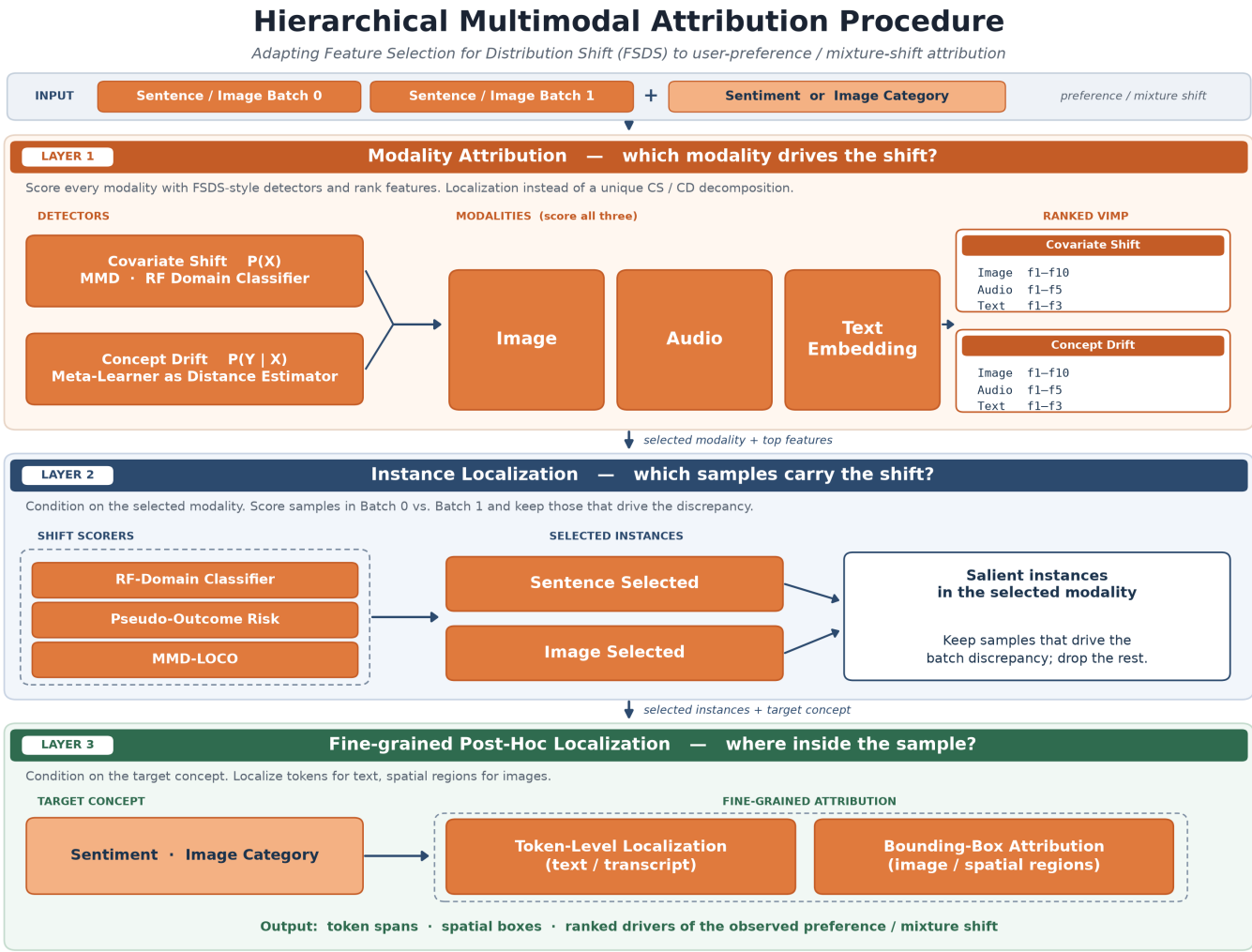
- Regarding of concept drift, the causal inference methods can be leveraged including permuCATE(concise feature selection) & causal forest split VIMP procedure(most parsimonious, relatively efficient) are recommended in practice.
- The causal inference method is designed to detect the equivalence of conditional mean $E[Y|X, T = 1]$, $E[Y|X, T = 0]$. The numerical studies and real-data analysis also demonstrated the SOTA performance of the causal inference methods in a wide variety of types of data.
- The software is open-sourced in https://github.com/ruanhq-StatsML/FSDS-FeatureSelection_for_DistributionShift.

RFPerm procedure for the LLM generated data

- The RFPerm procedure achieves desirable statistical power with as few as 30–40 observations.
- We propose RFPerm-as-Judge for continuous quality monitoring of LLM-generated synthetic data - its lightweight nature makes it ideally suited for testing for each new observation in streaming data pipelines.

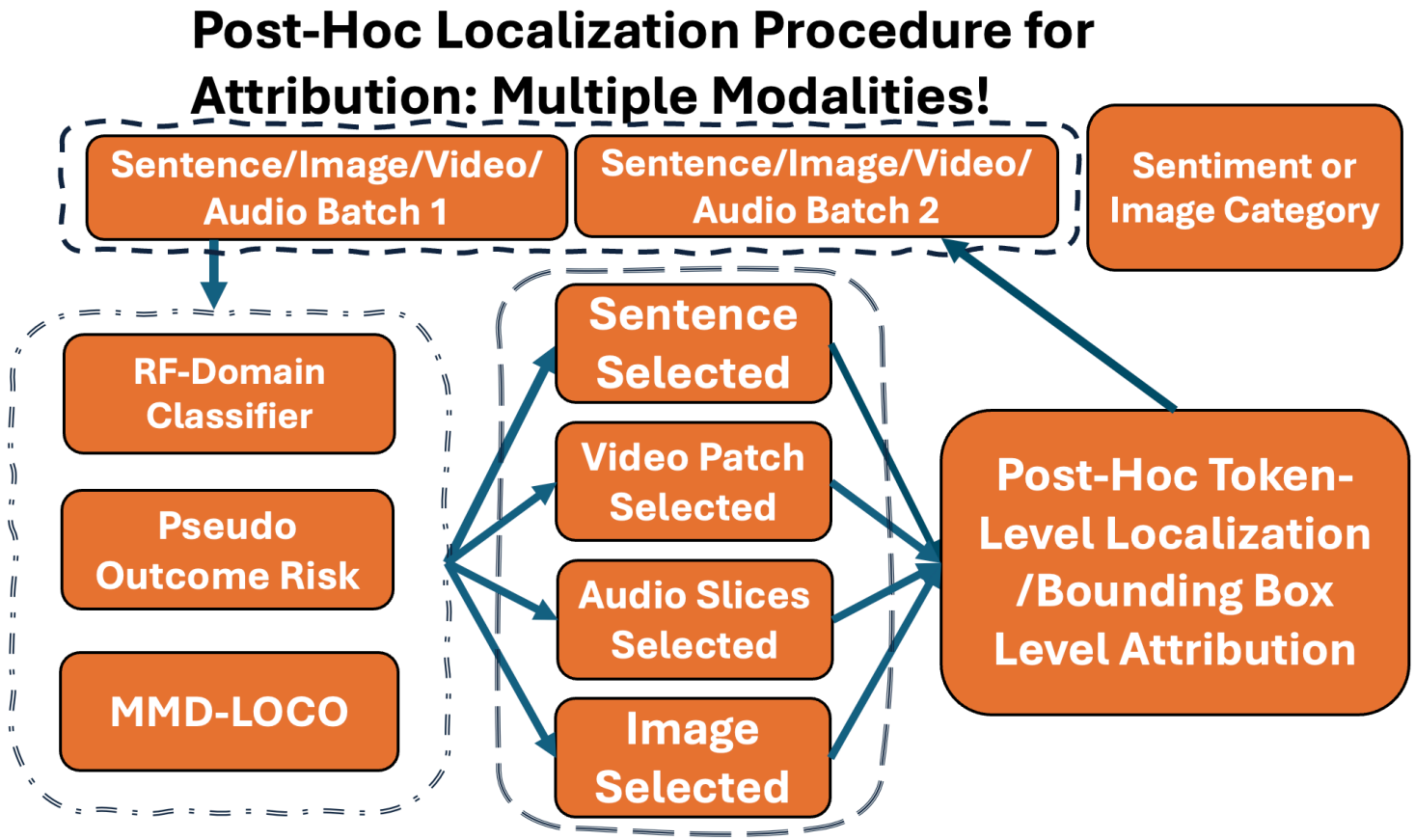
Multi-Modal Attribution

The Feature Selection for Distribution Shift procedure can be applied to the multi-modal attribution with multiple layers.



Post-Hoc Localization Followed by Attribution

Multi-Modal Attribution Followed by the post-hoc subset localization procedure.



Production Value: Adapting the Distribution Shift

- Once the significant distribution shift has been identified, the post-hoc subset localization procedure is highly preferred over attempting to uniformly disentangle the root causes across the entire population. This allows teams to focus their remediation efforts on the specific data segments where the degradation is most pronounced.
- To adapt to the model's performance drop introduced by the distribution shift, we can add density ratio, online rolling statistics, and localized propensity score as clever covariates - serve as shadow models. These serve as lightweight yet effective correction signals that can be injected directly into inference without full retraining.
- Meanwhile, the variable importance can be monitored along with confidence intervals via the online PFI procedure, providing stakeholders with an interpretable dashboard for real-time decision-making.

Production Value: Adapting the Distribution Shift

- The clever covariate procedure greatly enhances the interpretability of model behavior under distribution shift, while incurring very light computational cost — as it only augments inference-time features without requiring expensive model retraining or architectural changes.
- The post-hoc subset localization procedure delivers concise, actionable insights regarding why and how the distribution shift originates. Once the affected subset is identified, it naturally guides targeted clever covariate corrections, making the adaptation both precise and interpretable.
- The variable importance monitoring procedure can be further justified through trend tests or the localization diagnostics, which carry direct implications for feature knockout decisions — ensuring that a feature is deprecated only when its importance decay is sustained and statistically credible, rather than driven by transient noise.

Pseudo-outcome Risk

The Pseudo-Outcome Risk can serve as the quantification of the OOD(out-of-distribution) itself - as well as making contributions to multi-modal learning.

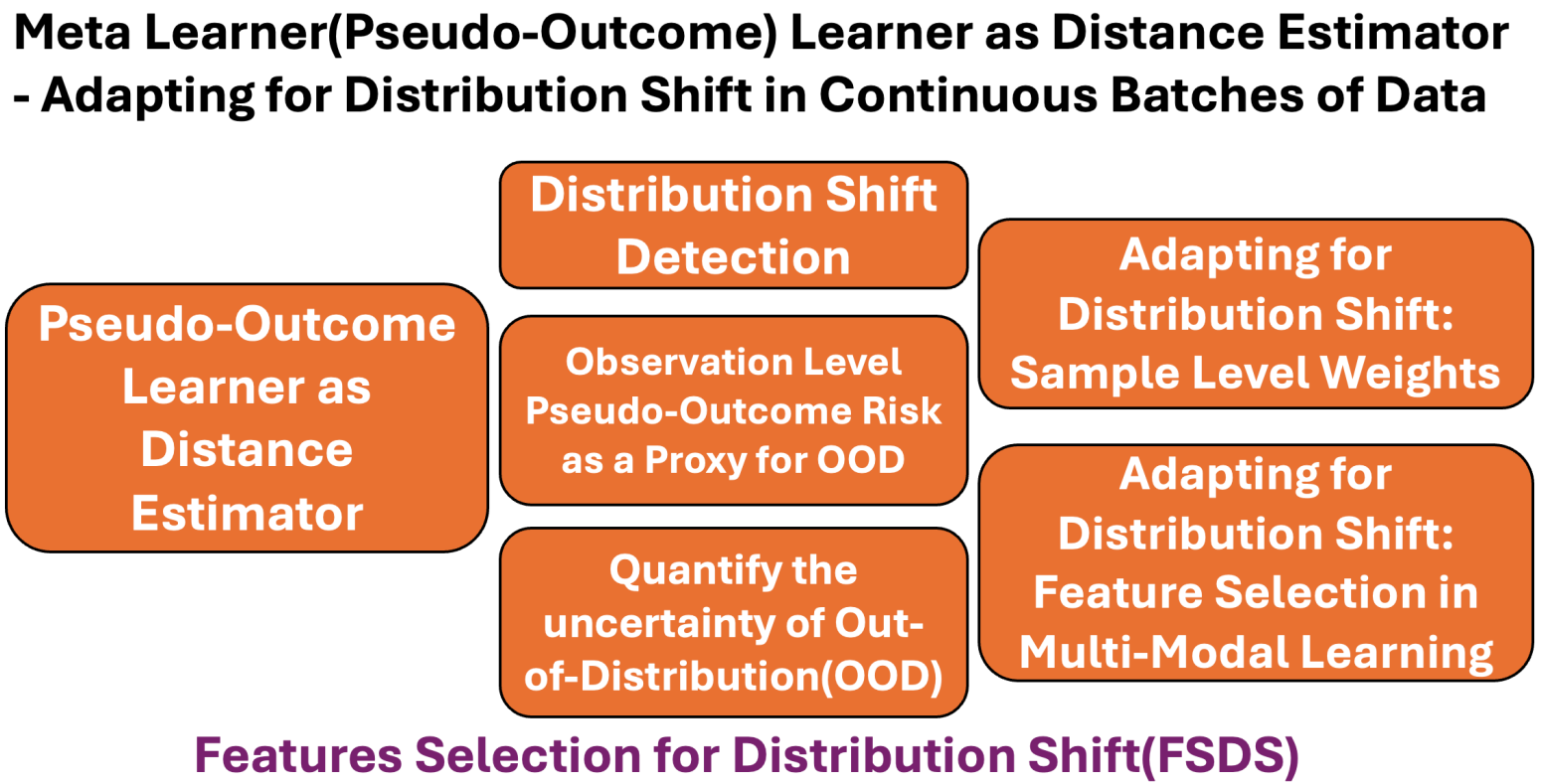


Illustration for Production Value

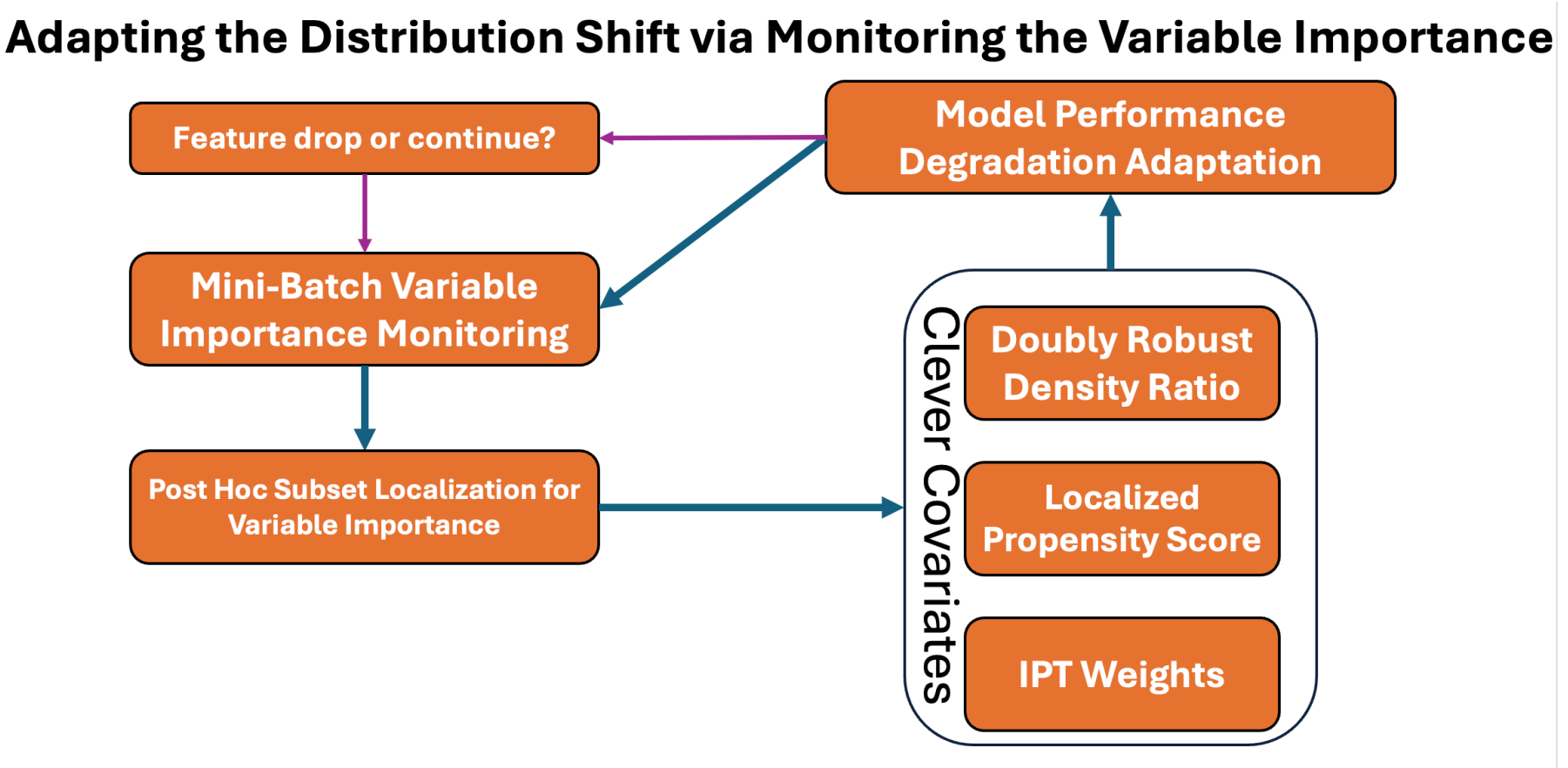


Figure: Overview of the Methodologies for adapting the distribution shift via the post hoc correction through the clever covariate

Production Value

Understanding distribution shift from feature level has direct practical value in the production ML System.

- The two-step procedure for the feature selection can enable the transformation from the hypothesis testing procedure to the feature selection mechanism.
- The covariate shift & concept drift attribution procedure, coupled with other model monitoring practices, can enable the near-real-time, automatic feature removal & granular root-cause diagnosis with the reduced MTTD/MTTA.

Production Value

Online Monitoring of Variable Importance - proactive monitoring

- Online incremental permutation importance – dynamically permutes feature order within a rolling window and records confidence intervals for performance degradation, providing a real-time measure of each feature's evolving contribution.
- Proactive decay monitoring alerting – automatically flags features that have become non-informative, triggering actionable alerts for ML scientists across both online streaming and offline mini-batch pipelines, drastically reducing reaction time to model drift.

Production Value

Integration into the Production System:

- The proposed feature selection procedure for distribution shift can be seamlessly integrated into existing model monitoring pipelines, complementing conventional metrics such as PSI, KL divergence, and the KS statistic in deployed ML systems.
- While fully assumption-less identification of distribution shift drivers is infeasible, the feature selection for distribution shift procedure is still capable of extracting a ranked list of the top-k features contributing significantly to the observed distribution shift, providing actionable references for Business, Operation and Engineering teams.

Production Value

This thesis propose two lines of Statistical Methodologies: Streaming Testing and Batch Testing procedures that is highly generalizable to a wide variety of production scenarios.

- Batch Testing: Parallelization available for Permutation Test, computationally feasible for conducting the test each 5 minutes.
- Production Landing for the Streaming Testing: Near real-time inference & hypothesis testing for the Streaming Observations to trigger Model Refresh Pre-Signals.
- Highly Flexible: all of these methodologies can be adapted to a wide variety of types of datasets & modalities.

Two use cases for the FSDS

- We leverage the FSDS procedure to enhance continuous and efficient collaboration in multi-agent systems, using the pseudo-outcome learner and random forest. The weaker the relationships between agents, the greater the marginal improvement the agents can provide.
- We leverage graph clustering procedure (like ST,spatial temporal _DBSCAN) followed by FSDS localization to enhance the identification of fraud patterns and behaviors. Graph-based information can then serve as a clever covariate in the next step to further adapt to model performance drops introduced by distribution shift.

Future Directions

- Generalize the clever covariate and multi-layer attribution procedure to multi-modal generative models, where modality-specific attribution results guide downstream training and inference.
- Leverage a hierarchical attribution procedure to enhance next-stage LLM inference and model training - for any kinds of LLM indeed.

Table: Production Use-Case for the onlineRFPerm - related to LLM

Features	Y	Significance means	Significance does not mean
Reasoning smoothness	Whether hop/path is valid	Reasoning chain broken	Hallucination rate
Review / judge	Pass / fail	Policy pack or judge swapped	Rejection rate; HH chosen
Graph-RAG subgraph	Citations on edges; support nodes in graph pack	Graph pack or community swapped	Pointwise relevance
Hybrid retrieval	Whether fused answer holds	One index or fusion broken	Single-path recall
Agent next step	Schema valid, tools callable, no loop	Protocol broken or spinning	Final task success
Synthetic data gold label	Whether still usable as observed samples	Synthetic distribution drifted	"Looks human-written"
Serving refresh moment	Whether current template + retrieval + generation still predicts success	Time to swap pack	Whether LLM needs fine-tuning
Experiment CUPED	Whether covariate adjustment still works	Time to redo regression	Which A/B arm won

Monitoring Uplift Models

Uplift model is widely deployed in ads, incentive modeling, budget allocation - the current post-deployment monitoring procedure is very scarce. The model fails silently under the batch shift or the user mixture shift - offline metrics on frozen holdout looks fine. $\hat{\tau}(\mathrm{x}) = \mathrm{f}(\hat{\mu}_1(\mathrm{x}), \hat{\mu}_0(\mathrm{x}))$.

- Outcome model $\hat{\mu}_\mathrm{t}$ — hard to monitor: label delay, counterfactual missing, selection bias, error cancellation / amplification.
- Propensity model $\hat{\mathrm{e}}$ — largely self-contained: calibration, distribution, extreme-value checks suffice.
- Core idea: Adapt the same batch FSDS stack; one change — AUUC replaces MSE as the ranking score.
- Setup: Existing batch (calibration) vs. New batch (deployment); batch label W, treatment T, outcome Y, features X. SRM on T gates validity.

When to update the uplift model?

The MMD, Pseudo-outcome Risk, and the AUUC, along with the post-hoc subset localization procedures, are adapted.

- PSM is self-contained: monitor calibration, distribution, and extreme values; refit directly when drift is detected.
- Outcome model requires caution: combine the AUUC gap with overall model performance—never rely on a single signal.
- Drift signals (MMD, PO-risk, the drop of AUUC) trigger the post-hoc localization to identify affected quantiles and corresponding feature groups.
- Update order: RefreshRef $\rightarrow$ Reallocate / NarrowSupport $\rightarrow$ Relearn only after trial failure; no immediate full retrain.

Developed Methods

The software implementation and accompanying demonstrations can be accessed via https://github.com/ruanhq-StatsML. For additional information and potential collaborations - please direct correspondence to ruanhqustc@outlook.com.